

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25

1. Title: Glyphosate-based herbicide enhances the uterine sensitivity to estradiol in rats

Authors: Marlise Guerrero Schimpf^{1,2}, María M. Milesi^{1,2}, Enrique H. Luque^{1,2}, Jorgelina Varayoud^{1,2*}

¹*Instituto de Salud y Ambiente del Litoral (ISAL, UNL-CONICET), Facultad de Bioquímica y Ciencias Biológicas, Universidad Nacional del Litoral, Santa Fe, Argentina.*
²*Cátedra de Fisiología Humana, Facultad de Bioquímica y Ciencias Biológicas, Universidad Nacional del Litoral, Santa Fe, Argentina.*

***Corresponding author:** Instituto de Salud y Ambiente del Litoral (ISAL), Facultad de Bioquímica y Ciencias Biológicas, Universidad Nacional del Litoral, Casilla de Correo 242, Santa Fe 3000, Argentina. Tel.: +54 342 4575207; fax: +54 342 4575207.
E-mail address: varayoud@fcb.unl.edu.ar (J. Varayoud).

Short title: Estradiol sensitivity after herbicide exposure.

Key words: estrogen receptors; uterus; neonatal; rat.

Word count of the full article: 6454 words.

26 **2. Abstract**

27 In a previous work, we detected that postnatal exposure to a glyphosate-based herbicide (GBH)
28 alters uterine development in prepubertal rats causing endometrial hyperplasia and increasing
29 cell proliferation. Our goal was to determine whether exposure to low-dose of a GBH during
30 postnatal development might enhance the sensitivity of the uterus to an estrogenic treatment.
31 Female Wistar pups were subcutaneously injected with saline solution (control) or GBH using
32 the reference dose (2 mg/kg/day, EPA) on postnatal days (PND) 1, 3, 5, and 7. At weaning
33 (PND21), female rats were bilaterally ovariectomized and treated with silastic capsules
34 containing 17 β -estradiol (E2, 1mg/ml) until they were two months of age. On PND60, uterine
35 samples were removed and processed for histology, immunohistochemistry and mRNA
36 extraction to evaluate: i) uterine morphology, ii) uterine cell proliferation by the detection of
37 Ki67, iii) the expression of the estrogen receptors alpha (ESR1) and beta (ESR2), and iv) the
38 expression of WNT7A and β -catenin. GBH-exposed animals showed increased luminal
39 epithelial height and stromal nuclei density. The luminal and glandular epithelium were
40 markedly hyperplastic in 43% of GBH-exposed animals. GBH exposure caused an increase in
41 E2-induced cell proliferation in association with an induction of both ESR1 and ESR2. GBH
42 treatment decreased membranous and cytoplasmic expression of β -catenin in luminal and
43 glandular epithelial cells and increased WNT7A expression in the luminal epithelium. These
44 results suggest that early postnatal exposure to a GBH enhances the sensitivity of the rat uterus
45 to estradiol, and induces histomorphological and molecular changes associated with uterine
46 hyperplasia.

47

48 **3. Introduction**

49 Estrogens have been implicated as important etiologic agents in cancer of the female
50 reproductive system (Chuffa et al. 2017). Likewise, several epidemiological studies have
51 demonstrated a close correlation between environmental and/or occupational exposure to
52 endocrine disrupting chemicals (EDCs) with estrogenic activity and cancer in women (Cohn et
53 al. 2015; Lerro et al. 2015). Experimental studies performed in rodents have shown that
54 environmental pollutants with estrogenic activity can induce pre-neoplastic uterine lesions as
55 well as endometrial cancer. Yoshizawa et al. (2009) performed an *in vivo* study where female
56 adult Harlan Sprague-Dawley rats orally exposed to chronic administration of dioxin and
57 dioxin-like compounds (polychlorinated biphenyls and furans) showed increased incidence of
58 cystic endometrial hyperplasia, luminal squamous metaplasia and carcinoma in the uterus. Other
59 authors demonstrated that developmental exposure to diethylstilbestrol (DES) and bisphenol A
60 (BPA) increases the incidence and severity of benign, premalignant and neoplastic uterine
61 lesions in aged hamsters, rats, and mice, including atypical hyperplasia and endometrial
62 adenocarcinoma (Leavitt et al. 1981; Newbold et al. 1990; Newbold et al. 2007; Vigizzi et al.
63 2015, 2016).

64 Among environmental pollutants, glyphosate-based herbicides (GBHs) have been one of the
65 most intensive used over the last two decades worldwide (Benbrook 2016), resulting in
66 increased environmental, plant and food contamination (Bai & Ogbourne 2016; Primost et al.
67 2017; Rodrigues & de Souza 2017; Zoller et al. 2017; Van Bruggen et al. 2018). Nonetheless,
68 estimated human exposure from glyphosate maximum residue levels measured in food items
69 and human urine samples has indicated that daily exposure to glyphosate is below the tolerable
70 reference doses currently established by regulatory agencies (Niemann et al. 2015; Solomon
71 2016; Zoller et al. 2017). However, these tolerable doses of glyphosate are being questioned
72 because they rely on outdated studies and may fail to protect human health or the environment
73 (Vandenberg et al. 2017).

74 In the last years, it has been controversy and debate regarding the carcinogenic and the
 75 endocrine disrupting potential of glyphosate and GBHs. Several bodies of experts have
 76 evaluated glyphosate carcinogenicity and drawn different conclusions. The WHO's
 77 International Agency for Research on Cancer working group (IARC) concluded "glyphosate is
 78 probably carcinogenic to humans (IARC Group 2A)" (IARC 2015), based on the evaluation of
 79 the publicly available evidence including data on the active ingredient glyphosate as well as on
 80 GBHs. In contrast, the European Food Safety Authority (EFSA) concluded "glyphosate is
 81 unlikely to pose a carcinogenic hazard to humans" (EFSA 2015), based on all available data on
 82 the active ingredient glyphosate, including industry data (Portier et al. 2016; Tarazona et al.
 83 2017). Similarly, different *in vitro* and *in vivo* studies have revealed possible endocrine-
 84 mediated effects of glyphosate and its commercial formulations on hormone-dependent tissues
 85 such as testis (Cassault-Meyer et al. 2014), ovary (Perego et al. 2017) and uterus (Guerrero
 86 Schimpf et al. 2017; Varayoud et al. 2017). Glyphosate was suggested to have endocrine
 87 disrupting properties by inhibiting aromatase activity (Richard et al. 2005) and activating the
 88 estrogen receptor alpha (ESR1) and beta (ESR2) in breast cancer cells (Thongprakaisang et al.
 89 2013; Mesnage et al. 2017). However, no evidence of potential interaction of glyphosate with
 90 endocrine pathways has been detected in the Endocrine Disruptor Screening Program (EDSP)
 91 conducted by the US Environmental Protection Agency (EPA) (US EPA 2015). Thus, the
 92 carcinogenic and endocrine disrupting potential of glyphosate and GBHs remains uncertain.

93 In a previous work, we demonstrated that the exposure to low dose (2 mg/kg of body weight
 94 (bw)/day) of a GBH during the first postnatal days (PND) alters the development of the rat
 95 uterus by affecting uterine morphology and the expression of proteins that regulate uterine
 96 organogenetic differentiation during the neonatal (PND8) and prepubertal (PND21) periods
 97 (Guerrero Schimpf et al. 2017). In fact, GBH induced endometrial hyperplasia in association
 98 with increased cell proliferation and deregulated different uterine morphoregulatory genes, such
 99 as ESR1, progesterone receptor (PR), homeobox A10 and wingless-type MMTV integration site
 100 family, member 7A (WNT7A) (Guerrero Schimpf et al. 2017). Alterations in uterine
 101 histoarchitecture and disruption of uterine morphoregulatory genes during critical periods of

development might lead to reproductive anomalies during adulthood, such as infertility and early pregnancy loss, and could promote the development of uterine neoplasias (Newbold et al. 1990; Newbold et al. 1997; Varayoud et al. 2008; Varayoud et al. 2014). In this sense, we detected that early postnatal exposure to GBH causes post-implantation embryo loss in adult female rats associated with an altered decidualization response and defective uterine differentiation/proliferation (Ingaramo et al. 2016, 2017). However, less evidence are available about the capability of GBHs to promote uterine neoplasias.

Based on controversial and limited studies, we evaluated whether exposure to low-dose GBH during early stage of development might enhance the sensitivity of the uterus to an estrogenic prepubertal treatment. Female rats postnatally exposed to vehicle or GBH were bilaterally ovariectomized at weaning (PND21) and submitted to a chronic estrogen stimulation until PND60 to evaluate the following end-points: i) uterine morphology, ii) uterine cell proliferation, iii) the expression of the estrogen receptors (ERs) (ESR1 and ESR2), and iv) the expression of WNT7A and β -catenin (*Ctnnb1*), selected as E2-modulated targets that have been implicated in altered uterine E2 responsiveness and neoplasia.

4. Materials and methods

4.1. Animals

All procedures used in this study were performed in accordance with the principles and procedures outlined in the Guide for the Care and Use of Laboratory Animals issued by the US National Academy of Sciences and were approved by the Institutional Ethics Committee of the School of Biochemistry and Biological Sciences (Universidad Nacional del Litoral, Santa Fe, Argentina). Inbred Wistar strain rats were bred at the Department of Human Physiology (Santa Fe, Argentina) and housed in a controlled environment (22°C $\pm$ 2°C; lights on from 06:00 to 20:00 h) in stainless steel cages with sterile pine wood shavings as bedding. Rats had free access to pellet laboratory chow (16-014007 Rat-Mouse Diet, Nutrición Animal, Santa Fe, Argentina) and tap water. For more information regarding the food composition, see: Kass et al. 2012, Supplementary data Table 1; and, Vigezzi et al. 2016.

130

131 **4.2. Experimental design**

132 Pups were obtained from timed-pregnant rats housed singly. After delivery (PND0), pups were
133 sexed according to anogenital distance and cross-fostered by distributing the pups of each litter
134 among the mothers. This procedure minimizes the use of siblings and thus avoids potential litter
135 effects. Cross-fostered litters were adjusted to 8 pups, with 8 female pups per litter. When fewer
136 than 8 females were available, an appropriate number of males were retained. Female pups from
137 each foster mother were randomly assigned to one of the following postnatal treatment group:
138 1) control group receiving saline solution, and 2) GBH group receiving a commercial
139 formulation of glyphosate dissolved in saline solution (2 mg of glyphosate/kg bw). The
140 glyphosate formulation used was Roundup FULL II[®], a liquid water-soluble formulation
141 containing 66.2% of glyphosate potassium salt (equivalent to 54% w/v of glyphosate acid), as
142 its active ingredient, coadjuvants and inert ingredients. Forty microliters were administered by
143 subcutaneous (sc) injection in the nape of the neck every 48 h from PND1 to PND7. Each
144 treatment day, the dose was calculated based on the concentration of glyphosate acid (54 g of
145 glyphosate per 100 ml of GBH) and the average body weight of the pups. The dose of GBH
146 selected to perform this study is in the order of magnitude of the reference dose (RfD) of 1
147 mg/kg bw/day recently reassigned for glyphosate by the US EPA (2017). Moreover, this dose is
148 representative of the glyphosate residues found in soybean grains (Arregui et al. 2004; Test
149 Biotech, 2013), and is in the order of magnitude of the environmental levels detected in our
150 country (Peruzzo et al. 2008; Bonansea et al. 2017; Primost et al. 2017). Although the RfD for
151 glyphosate is based on oral exposure, the subcutaneous administration enhances delivery of
152 chemical compounds in an early postnatal exposure model (Milesi et al. 2012). The early
153 postnatal model of exposure to endocrine disruptors has been extensively used in our laboratory
154 in both rodents (Ramos et al. 2007, Varayoud et al. 2008; Monje et al. 2009; Milesi et al. 2015;
155 Ingaramo et al. 2016, 2017; Guerrero Schimpf et al. 2017) and lambs (Rivera et al. 2011) and
156 has been demonstrated as a persuasive paradigm to study short- and long-term consequences of
157 neonatal exposure to hormonally active substances (Rivera et al. 2015). As previously reported

(Guerrero Schimpf et al. 2017), GBH postnatal treatment (with 2 mg/kg bw/day in pups every 48 h from PND1 to PND7) produced no alterations in maternal care and nursing between the experimental groups. No signs of acute or chronic toxicity were observed in the litters, and no differences in weight gain between GBH and control pups were recorded during the experiment. A schematic representation of the experimental design is shown in Fig. 1.

4.3. Ovariectomy, estrogen treatment and sample collection

To study the long-term effects of postnatal GBH exposure on uterine sensitivity to E2, we used a model of prepubertal ovariectomized (OVX) rats submitted to chronic estrogen treatment (Fig. 1). At weaning (PND21), control (C) and postnatally GBH-exposed female rats were anesthetized with a solution of ketamine:xylazine:acepromazine, 50:5:1 mg/ml. The solution was prepared by mixing 1.0 ml of ketamine hydrochloride (50 mg/ml, Ketamina 50, Holliday-Scott, Buenos Aires, Argentina), 0.25 ml of xylazine hydrochloride (20 mg/ml, Kensol® König, Buenos Aires, Argentina) and 0.1 ml of acepromazine maleate (10 mg/ml, Acedan, Holliday-Scott, Buenos Aires, Argentina). The anesthetic solution was administered intraperitoneally at a dose of 135 μ l/100 g bw. Animals were bilaterally OVX to avoid endogenous estradiol (E2) variability. A group of control (C+OVX+E2, n=8) and GBH-exposed (GBH+OVX+E2, n=7) females were submitted to a chronic estrogen treatment, using silastic capsules (inner/outer diameter: 1.98/3.18 mm, 19 mm in length, Dow Corning Corp., Midland, MI, USA) containing 1 mg of 17 β -estradiol/ml (Sigma-Aldrich, St. Louis, MO, USA) dissolved in corn oil. The capsules were preincubated at 37°C in phosphate buffer for 24 h to stabilize the release of E2, and then were implanted (one per animal) subcutaneously between the shoulder blades during the anesthesia used at ovariectomy. In order to characterize the uterine response to exogenous E2 treatment, we performed a pilot experiment using OVX prepubertal rats that were not exposed to GBH. For that purpose, another set of control females was implanted with capsules filled with vehicle alone (corn oil) (C+OVX+V, n=5), and the effects were compared against the control group submitted to E2 treatment (C+OVX+E2). The treatment with the E2 implants produces blood levels of E2 that are equivalent to E2 circulating levels during the rat estrous

cycle (Smith et al. 1975; Bottner & Wuttke 2005). This was verified by measuring E2 serum levels (see Section 4.4). After surgery, animals were allowed to recover in a clean cage under a heating lamp. Analgesia (2 mg/kg of flunixin meglumine, Aplonal[®], König, Buenos Aires, Argentina) was administered subcutaneously once per day and for three days, starting after capsule implantation. Once the animals were fully conscious and active, they were housed individually and daily monitored for signs of pain or distress for a period of one week. Female rats were sacrificed when they were two months of age (PND60) to collect trunk blood and uterine tissue. Blood samples were centrifuged (3000 rpm, 20 min) and serum was stored at -20 °C until hormone assay was performed. One uterine horn of each rat was fixed by immersion in 10% buffered formalin for 6 h at 4°C and processed for histology and immunohistochemistry (IHC). The other uterine horn of each rat was immediately frozen in liquid nitrogen and stored at -80 °C for RNA extraction followed by reverse transcription-polymerase chain reaction (RT-PCR) assay.

199

200 **4.4. Determination of E2 serum levels**

Serum samples stored at -20 °C were thawed and E2 serum levels were assessed using the Ultra-Sensitive Estradiol Radioimmunoassay Kit DSL4800 (Immunotech, Beckman Coulter, Czech Republic) according to the manufacturer's guidelines. All samples were run in duplicate. The assay sensitivity was 2.2 pg/ml. The intra and interassay coefficients of variation were ≤ 8.9 and 12.2 %.

206

207 **4.5. Histology and morphometry**

Uterine samples embedded in paraffin were longitudinally cut (5 μ m thick) at three different depths (separated 50 μ m from each other), mounted on slides coated with 3-aminopropyl triethoxysilane (Sigma-Aldrich), stained with hematoxylin and eosin (H&E) and examined by light microscope (Olympus BH2 microscope; Olympus, Tokyo, Japan) to analyze uterine morphology. Uterine sections were evaluated for histologic changes by a trained pathologist blinded to the experimental group. We also determined glandular density, stromal nuclei

density, luminal and glandular epithelial heights, and the thickness of the stroma and myometrium. To determine the glandular density, the volume fraction (V_v) of uterine glands was calculated by applying the following formula given by Weibel (1969): $V_v = P_i/P$, where V_v is the estimated volume fraction of the object (glands), P_i is the number of incident points over the glands, and P is the number of incident points over all cells in the studied population (stroma). To obtain the data for this point-counting procedure, a glass disk with a squared grid of 0.8 mm $\times$ 0.8 mm was inserted into a Dplan 40 $\times$ focusing eyepiece (numerical aperture = 0.65, Olympus) (Gundersen et al. 1988; Ramos et al. 2002). The results were expressed as $V_v \times 100$. The V_v of uterine glands was quantified on at least 20 randomly selected fields per section, and three sections per animal (separated 50 μ m from each other) were evaluated. The stromal nuclei density was defined as the ratio between the area occupied by stromal nuclei and total subepithelial stromal area (150- μ m-wide area adjacent to the epithelium, from the basement membrane toward the outer layer). Areas were quantified using Fiji of Image J, excluding uterine glands, infiltrating cells and blood vessel cells from analysis (Durando et al. 2007). The luminal and glandular epithelial heights, and the thickness of the stromal and myometrial (circular and longitudinal) layers were analyzed by image analysis using Fiji of Image J as previously described (Ramos et al. 2002; Varayoud et al. 2017). Briefly, the images were recorded with a Spot Insight V3.5 color video camera, attached to a microscope (Olympus). To spatially calibrate the software analyzer, square grids from Neubauer's chamber images were captured. At least five fields were recorded in each section using a Dplan 20 $\times$ objective (numerical aperture = 0.40, Olympus), and three sections per animal (separated 50 μ m from each other) were evaluated.

236

237 **4.6. Reverse transcription and real-time quantitative PCR analysis (qRT-PCR)**

238 **4.6.1. RNA extraction and reverse transcription**

239 Individual uterine horn samples from each experimental group were homogenized in TRIzol[®] reagent (Invitrogen, Carlsbad, CA, USA) and total RNA was extracted according to the manufacturer's instructions. The concentration and purity of total RNA was determined by

measuring the optical density at 260 and 280 nm. All samples were precipitated with ethanol, dissolved in distilled water, and then stored at -80°C until needed. Equal quantities (1 μg) of total RNA were reverse-transcribed into cDNA with Moloney Murine Leukemia Virus reverse transcriptase (MMLV-RT; 300 units; Promega, Madison, WI, USA) using 200 pmol of random primers (Biodynamics, Buenos Aires, Argentina). Twenty units of ribonuclease inhibitor (RNAout, Invitrogen Argentina, Buenos Aires, Argentina) and 100 nmol of a deoxynucleotide triphosphate mixture were added to each reaction tube at a final volume of 30 μl of 1 $\times$ MMLV-RT buffer. Reverse transcription was performed at 37°C for 90 min and at 42°C for 15 min. Reactions were stopped by heating at 80°C for 5 min and cooling on ice. Each reverse-transcribed product was diluted with RNase-free water to a final volume of 60 μl .

4.6.2. qRT-PCR

The mRNA expression levels of total *Esr1*, *Esr2*, *Wnt7a*, *Ctnnb1* and ribosomal protein L19 (*Rpl19*, housekeeping gene) were quantified by qRT-PCR using the Rotor-Gene Q cycler (Quiagen; Tecnolab; Buenos Aires, Argentina). The gene-specific primer sequences are shown in Table 1. For cDNA amplification, 5 μl of cDNA were combined with HOT FIREPol[®] EvaGreen[®] qPCR Mix Plus (Solis BioDyne; Biocientífica, Rosario, Argentina) and 10 pmol of each primer (Invitrogen) in a final volume of 20 μl . After initial denaturation at 95°C for 15 min, the reaction mixture was subjected to successive cycles of denaturation at 95°C for 15 s, annealing at $56-60^{\circ}\text{C}$ for 15 s, and extension at 72°C for 15 s. Product purity was confirmed by dissociation curves and random samples were subjected to agarose gel electrophoresis. Controls containing no template DNA were included in all assays, yielding no consistent amplification. The cycle threshold (C_T) for each sample was calculated using the Rotor-Gene Q Series Software (Version 1.7, Quiagen; Tecnolab) with an automatic fluorescence threshold setting. The efficiency of PCR reactions was assessed for each target by the amplification of serial dilutions (over five orders of magnitude) of cDNA fragments of the transcripts under analysis. Accordingly, the fold expression over control values was calculated for each target by the relative standard curve methods, which are designed to analyze data from real-time PCR (Cikos

et al., 2007). For all experimental samples, the target quantity is determined from the standard curve, normalized to the quantity of the housekeeping gene and finally divided by the target quantity of the control sample. No significant differences in C_T values were observed for *Rpl19* among the experimental groups.

4.7. Immunohistochemistry

The protein expression of the cell proliferation marker Ki67 and the ERs (ESR1 and ESR2) was analyzed to evaluate the uterine response to exogenous E2 treatment. WNT7A and β -catenin protein immunostaining was evaluated as E2-modulated targets that have been implicated in altered uterine E2 responsiveness and neoplasia. A standard immunohistochemical technique was performed following protocols previously described by our laboratory (Muñoz-de-Toro et al. 1998). Briefly, uterine longitudinal sections (5 μ m thick) were deparaffinized and rehydrated in graded ethanol solutions. After microwave pretreatment for antigen retrieval, the endogenous peroxidase activity and non-specific binding sites were blocked. Samples were incubated in a humid chamber with the specific primary antibody (overnight at 4°C) and then with the corresponding biotin-conjugated secondary antibody (30 min at room temperature) (described in Table 2). Reactions were developed using the avidin-biotin-peroxidase method and diaminobenzidine (Sigma-Aldrich) as a chromogen substrate. Each immunohistochemical run included negative controls in which the primary antibody was replaced by non-immune horse serum (Sigma-Aldrich). For Ki67 immunodetection, samples were counterstained with Mayer's hematoxylin (Biopur, Rosario, Argentina). Samples were dehydrated and mounted with permanent mounting medium (Eukitt, Sigma-Aldrich).

4.8. Quantification of cell proliferation

Tissue sections (three per animal, separated 50 μ m from each other) were evaluated using the Olympus BH2 microscope with the Dplan 40 $\times$ objective (numerical aperture = 0.65; Olympus). The proliferation rate was assessed in the luminal and glandular epithelium as a percentage of Ki67-positive cells on a total of 2000 cells per compartment in each uterine section. In the

subepithelial stroma, the proliferation rate was obtained considering the Vv of the Ki67-positive cells, as previously described in section 4.5 of Materials and Methods.

300

301 **4.9. Quantification of protein expression by image analysis**

The protein expression of the ERs (ESR1 and ESR2), WNT7A and β -catenin in uterine cells was evaluated by image analysis, using Fiji of Image J as previously described (Ramos et al. 2002). In brief, immunostained images were captured with a Dplan 40 $\times$ objective (numerical aperture =0.65; Olympus) attached to a Spot Insight V3.5 color video camera. After conversion of each image into a gray scale, the integrated optical density (IOD) was measured as a linear combination of the average gray intensity and the relative area occupied by the positive cells (Ramos, et al. 2001; Ramos et al. 2002). Because the IOD is a dimensionless parameter, the results were expressed as arbitrary units. Protein quantification was performed in duplicate in the luminal and glandular epithelium, and in the subepithelial stroma (150- μ m-wide area adjacent to the epithelium, from the basement membrane toward the outer layer) on two uterine sections, separated 50 μ m from each other. Moreover, in each uterine section quantification was performed on at least 10 randomly selected fields.

314

315 **4.10. Statistical analysis**

Results are expressed as the mean $\pm$ SEM. All data were analyzed using a Mann-Whitney *U* test (GraphPad Prism Version 5.03). Values with $p < 0.05$ were accepted as significant.

318

319 **5. Results**

320 **5.1. Estradiol serum levels**

In order to know the serum levels of 17 β -estradiol reached by E2 implants, the concentration of E2 was measured in the serum of C+OVX+V, C+OVX+E2 and GBH+OVX+E2 female rats. As expected, E2 serum levels were below the detection limit of the assay in OVX rats treated with vehicle (C+OVX+V group). In C+OVX+E2 females, E2 implants raised E2 serum concentrations to around 30 pg/ml. No differences were observed in control and postnatally

326 GBH-exposed animals treated with E2 (C+OVX+E2: 33.1 ± 3.3 pg/ml; GBH+OVX+E2: $31.6 \pm$
 327 2.0 pg/ml, $p=0.90$) (Table 3).

328

329 **5.2. Characterization of uterine responses to E2 treatment**

330 Our first aim was to characterize the uterine response to estrogenic treatment between
 331 C+OVX+V and C+OVX+E2 by comparing uterine morphology, cell proliferation and the
 332 expression of both ESR1 and ESR2. As expected, uterine morphology was clearly different
 333 between vehicle- and E2-stimulated rats. The larger longitudinal section area of the E2-exposed
 334 uteri was evident at low magnification (Fig. 2A vs 2B). Uteri from C+OVX+V females revealed
 335 an atrophic endometrium with simple cuboidal epithelium in the lumen and glands as well as
 336 densely-packed stromal cells (Fig. 2C). In contrast, the luminal epithelium in E2-exposed uteri
 337 (C+OVX+E2 group) was hypertrophic (pseudostratified columnar cells) to low-grade
 338 hyperplastic. Additionally, the subepithelial stroma showed signs of water imbibition and was
 339 heavily infiltrated by inflammatory leukocytes (Fig. 2D).

340 Uterine response to E2 involves the differential modulation of the ERs and changes in cell
 341 proliferation. In our model of exposure, E2 administration resulted in downregulation of *Esr1*
 342 mRNA levels (Fig. 3A; $p<0.05$) and upregulation of the *Esr2* mRNA levels (Fig. 3B; $p<0.05$).
 343 Immunohistochemical analysis of protein expression showed that E2-treated animals exhibited
 344 down-regulation of ESR1 (Fig. 3C and D vs. E; $p<0.05$) and upregulation of ESR2 (Fig. 3C and
 345 F vs. G; $p<0.05$) in all uterine regions evaluated. In addition, a dramatic increase in cell
 346 proliferation in both luminal and glandular epithelial cells was detected after E2 treatment (Fig.
 347 3C and H vs. I; $p<0.01$).

348

349 **5.3. Long-lasting effects of GBH on uterine sensitivity to E2**

350 **5.3.1. Uterine histomorphology**

351 Histological features of uterine sections are illustrated in Fig. 4. As previously mentioned, E2
 352 stimulation prompted hypertrophy (pseudostratified columnar cells) and low-grade hyperplasia
 353 in the uterine luminal epithelium (C+OVX+E2 group) (Fig. 4A and D). We observed that uteri

of female rats postnatally exposed to GBH and treated with E2 (GBH+OVX+E2 group) showed histomorphological alterations. In fact, the luminal and glandular epithelium were markedly hyperplastic (43% of animals). GBH+OVX+E2 rats showed areas of endometrium with stratification of the luminal epithelium, hypochromic nuclei and mitotic figures (Fig. 4B and E). Furthermore, in GBH-exposed uteri treated with E2, several abnormal glands were observed that contained cells that were morphologically similar to those of the luminal epithelium (Fig. 4C and F). In addition, GBH-exposed animals showed increased luminal epithelial height and stromal nuclei density (Table 4). The evaluation of other morphological parameters did not show differences between control and postnatally GBH-exposed animals (C+OVX+E2 vs. GBH+OVX+E2) (Table 4).

5.3.2. Uterine proliferation

Evaluation of cell proliferation was performed by the assessment of Ki67 expression. An increased proliferative response to E2 treatment was detected in the postnatally GBH-exposed uteri. The evaluation of cell proliferation in the luminal epithelial compartment indicated that in GBH+OVX+E2 females the Ki67 expression was higher ($29.07 \pm 3.89\%$, $p < 0.05$) than in the C+OVX+E2 group ($16.36 \pm 1.10\%$) (Fig. 5A). Photomicrographs illustrating these results are shown in Fig. 5B. Glandular epithelial cells showed similar changes, but the induction of cell proliferation was not different between control and GBH-exposed females (C+OVX+E2: $16.32 \pm 1.42\%$; GBH+OVX+E2: $20.16 \pm 3.42\%$, $p = 0.62$) (Fig. 5A). Finally, the stromal cell proliferation was very low showing no changes between C+OVX+E2 and GBH+OVX+E2 treated animals (data not shown).

5.3.3. Expression of the estrogen receptors

Uterine expression of the ESR1 and ESR2 in control and GBH-exposed animals that received E2 treatment was evaluated at mRNA and protein levels. As previously mentioned, E2 stimulation induced downregulation of *Esr1* mRNA (C+OVX+E2 vs. C+OVX+V rats, Fig. 3A). We found that when rats were postnatally exposed to GBH, the *Esr1* mRNA downregulation

382 was accentuated (Fig. 6A; $p<0.05$). Likewise, the induction of *Esr2* mRNA detected in
 383 C+OVX+E2 animals, was enhanced in the uteri from GBH+OVX+E2 group (Fig. 6B; $p<0.05$).
 384 A similar response was detected when we analyzed the protein expression of ESR2. Postnatal
 385 exposure to GBH enhanced the ESR2 increase caused by E2 treatment in all uterine regions
 386 evaluated (luminal, glandular and stromal) (Fig. 6C; $p<0.05$). For ESR1, a strong induction was
 387 detected in the luminal epithelial cells of GBH-exposed animals (Fig. 6C, $p<0.05$), without
 388 changes in the subepithelial stroma or the glandular epithelium. Based on microscopic
 389 examination, increased levels of epithelial ESR1 expression were related not only to an
 390 increased in cell number but also to changes in expression levels within cells (the intensity of
 391 immunostaining). Representative images illustrating these results are shown in Fig. 6D.

392

393 **5.3.4. Expression of E2-modulated targets that have been implicated in altered uterine E2** 394 **responsiveness and neoplasia**

395 Uterine expression of WNT7A and β -catenin in control and GBH-exposed animals that received
 396 E2 treatment was evaluated at mRNA and protein levels. Treatment with GBH increased the
 397 expression of *Wnt7a* mRNA relative to control rats (Fig. 7A; $p<0.05$), while induced
 398 downregulation of *Ctnnb1* mRNA levels (Fig. 7B; $p<0.05$). Considering the differential role of
 399 these molecules according to their subcellular localization, we evaluated the spatial pattern of
 400 WNT7A and β -catenin expression in uteri of control and GBH-exposed rats by IHC. Both
 401 groups displayed similar expression patterns. WNT7A showed nuclear expression in the
 402 subepithelial stroma, and nuclear and cytoplasmic expression in uterine epithelial (luminal and
 403 glandular) cells. β -catenin displayed membranous and cytoplasmic pattern in the luminal and
 404 glandular epithelium with negative immunoreaction in the stroma. Despite no changes in the
 405 spatial pattern of WNT7A and β -catenin were observed, differences in immunostaining intensity
 406 were found between the experimental groups. Female rats postnatally exposed to GBH
 407 exhibited induction of WNT7A in the luminal epithelium (Fig. 7C; $p<0.05$). Surprisingly, β -
 408 catenin expression was notably decreased in the luminal and glandular epithelium of GBH-

409 treated animals (Fig. 7C; $p < 0.05$). Representative images illustrating these results are shown in
 410 Fig. 7D.

411

412 **6. Discussion**

413 Prior data obtained from siblings of the animals used in the present study revealed that the
 414 exposure to low dose (2mg/kg b.w/day) of a GBH during the first week of life resulted in
 415 alterations of uterine organogenetic and functional differentiation that caused subfertility later in
 416 life. In this line, GBH exposure elicited endometrial hyperplasia in prepubertal rats together
 417 with increased cell proliferation and altered expression of proteins involved in uterine
 418 organogenetic differentiation (Guerrero Schimpf et al. 2017). Long-term uterine effects included
 419 alterations in endocrine-dependent mechanisms involved in decidualization that led to post-
 420 implantation embryo loss. These alterations encompassed a dysregulation of ESR1-PR signaling
 421 (Ingaramo et al. 2016) and Wnt pathways (Ingaramo et al. 2017).

422

423 In the present study, we used a model in which animals were OVX on PND21 (prepubertal
 424 period). We hypothesized that GBH exposure during early stage of development might enhance
 425 the sensitivity of the rat uterus to E2. To test this hypothesis, different end-points were
 426 evaluated to determine the long-term uterine response to E2: i) uterine morphology, ii) uterine
 427 cell proliferation, iii) the expression of ESR1 and ESR2, and iv) the expression of WNT7A and
 428 β -catenin, selected as E2-modulated targets that have been implicated in altered uterine E2
 429 responsiveness and neoplasia.

430

431 The uterine responsiveness to E2 stimulation in OVX rodents has been extensively studied
 432 using different models. We chose chronic E2 treatment since it has been shown that prolonged
 433 E2 stimulation plays an important role in the development of endometrial pathologies such as
 434 hyperplasia and carcinoma (Chufa et al. 2010). In this model, serum concentrations of E2 in E2-
 435 implanted rats were equivalent to circulating levels during the rat estrous cycle (Smith et al.
 436 1975). No differences in E2 serum levels between control and GBH-exposed rats were detected,

437 indicating that all animals were exposed to the same levels of E2. Some studies have shown that
438 GBH exposure can disrupt aromatase (Richard et al. 2005; Cassault-Meyer et al. 2014; Defarge
439 et al. 2016), the key enzyme responsible for the conversion of androgens into estrogens. It is
440 worth noting here that, since we worked with OVX rats that received exogenous E2 treatment,
441 the contribution of extragonadal aromatization to E2 serum levels would be negligible.
442 However, taking into account that aromatase is an estrogen sensitive-gene and is related to
443 uterine disorders (Bulun et al. 2005), we cannot discard that aromatase expression could be
444 affected in GBH-exposed animals.

445

446 Using this model, we detected that when OVX animals were chronically exposed to E2, uterine
447 histomorphology showed signs of higher sensitivity to E2 in the postnatally GBH-treated group.
448 The results showed that the luminal and glandular epithelium were markedly hyperplastic.
449 Different morphological changes were detected in GBH-exposed animal, such as increase of
450 luminal epithelial height and higher stromal nuclei density, both histological features of simple
451 endometrial hyperplasia (Dixon et al. 2014; Sanderson et al. 2017). Similar alterations in both
452 uterine regions were observed at the neonatal period (PND8) (Guerrero Schimpf et al. 2017),
453 suggesting that GBH exposure permanently alters uterine histoarchitecture. Interestingly, DES-
454 exposed mice and hamsters that exhibited endometrial hyperplasia (Hendry et al. 1997;
455 Yamashita et al. 2006) developed uterine adenocarcinoma later in life (Leavitt et al. 1981;
456 Newbold et al. 1990). Considering that endometrial hyperplasia has the potential to progress to
457 endometrial cancer (Sanderson et al. 2017), our present findings provide the first evidence of a
458 link between developmental exposure to GBH and uterine development of pre-neoplastic
459 lesions after prolonged E2 exposure.

460

461 In addition, we observed a dramatic increase in uterine epithelial cell proliferation in association
462 with a strong induction of ESR1 in postnatally GBH-exposed females. ESR1 plays a crucial role
463 in uterine cell proliferation stimulated by E2. *In vivo* studies have demonstrated that both
464 stromal and epithelial ESR1 expression is important to induce uterine proliferation and

epithelial cell growth. Winuthayanon and colleagues (2014) have shown that epithelial ESR1 is essential for long-term uterine responsiveness to E2. Its expression induces gene products involved in mitosis and cell cycle progression as well as inhibition of apoptosis. These authors suggest that epithelial ESR1 may generate autocrine signals or provide a positive feedback to stromal cells to stimulate epithelial cell proliferation (Winuthayanon et al. 2010, 2014). Hence, the epithelial hyperplasia of GBH-exposed animals could be attributed, at least in part, to an increased expression of epithelial ESR1 that generates a maximal mitogenic signal. Moreover, our results are in concordance with the findings of Tica et al. (2016) who showed that ESR1 expression is increased in normal proliferative and benign hyperplastic epithelial cells. It is worth noting the differences in responsiveness to estrogens in relation to proliferative activity between luminal and glandular epithelial cells. In fact, luminal epithelial cells in GBH-exposed animals were more sensitive to E2-induced cell proliferation. Taking into account the important role of ESR1 over cell proliferation control, higher luminal epithelial proliferation might be related to higher ESR1 expression in this compartment. Contrary, in the glandular epithelium neither proliferation nor ESR1 expression were affected.

As previously mentioned, in our model of exposure, E2 administration induced downregulation of *Esr1* mRNA levels. In addition, GBH-exposed animals showed decreased expression of *Esr1* mRNA. However, a similar decrease in ESR1 protein expression was not observed. In fact, GBH treatment increased ESR1 in the luminal epithelium. In this context, we propose that GBH could affect protein turnover due to lower activity of the ubiquitin-proteasome pathway. It has been reported that proteasome-mediated proteolysis modulates the cellular concentration of ESR1 in a process that requires ligand binding to the receptor (Preisler-Mashek et al. 2002). Moreover, it has been demonstrated that the extent to which the overall ESR1 levels are affected depends on the ligand and is not related to ligand-binding affinity or activation of transcription (Preisler-Mashek et al. 2002).

ESR2 is important for differentiation and growth of the uterine epithelium. In addition, loss of ESR2 function results in an enhanced uterine responsiveness to E2 in the neonatal and adult uterus, highlighting its role in the inhibition of epithelial cell proliferation (Wada-Hiraike et al. 2006; Nakayima et al. 2015). Besides this protective action in normal endometrial tissue, a completely different role has been proposed in benign and malignant proliferative endometrial pathologies. Increased levels of ESR2 found in high-grade and advanced stage endometrial carcinomas as well as in endometriotic stromal cells suggest an involvement in aberrant cell proliferation and tumor-promoting activity (Bulun et al. 2010; Hapangama et al. 2015). Here, we found an increase in ESR2 uterine expression at mRNA and protein levels in GBH-exposed uteri after E2 administration. These results support the concept that alterations in ESR2 expression are important in the development and progression of uterine proliferative lesions.

In the endometrium, Wnt/ β -catenin signaling pathway is implicated in different processes such as uterine development and epithelial cell proliferation in response to E2 (Van der Horst et al. 2012). β -catenin plays a dual role as an adhesion molecule in adherens junctions at the plasma membrane and as a key intermediate in the canonical Wnt signaling pathway (Brembeck et al. 2006). WNT7A is a member of the Wnt protein family that is able to directly stimulate canonical Wnt signaling. We were interested in examining WNT7A expression because previous studies have shown that this protein is up regulated by E2 (Wang et al. 2009) and overexpressed in uterine disorders such as endometriosis and endometrial cancer (Gaetje et al. 2007; Liu et al. 2013). In our model, GBH treatment increased WNT7A mRNA and protein expression in the luminal epithelium, suggesting enhanced Wnt/ β -catenin signaling. However, the pattern of changes in β -catenin differed from those in WNT7A expression. Decreased membranous and cytoplasmic β -catenin expression without nuclear immunoreaction was detected in the luminal and glandular epithelium of GBH-exposed rats. In accordance with these results, decreased β -catenin expression was observed in epithelial cells from hyperplastic uterine lesions in mice treated with lithium and E2 (Gunin et al. 2004). Despite nuclear β -catenin accumulation is a well-known alteration in uterine hyperplasia and neoplasia, it has been

520 demonstrated that β -catenin cell membrane expression is decreased in these types of lesions
521 (Saegusa et al. 2001; Florescu et al. 2016). Loss of β -catenin expression on the cell surface
522 destabilizes and weakens cell-cell adhesion favoring cell dissociation (Guarino et al. 2007).
523 Taking into account morphologic changes induced by GBH-exposure, alterations in intercellular
524 contacts and continuous activation of Wnt/ β -catenin signaling pathway may contribute to the
525 development of endometrial cancer.

526

527 Few studies have shown that glyphosate and its formulations can interact with the ERs. Gasnier
528 et al. (2009) showed inhibition of ESR1 and ESR2 transcriptional activities by different GBH
529 formulations in human liver HepG2 cells transfected with an estrogen response element
530 construct and the ERs. On the contrary, other studies showed that glyphosate (the active
531 ingredient of formulations) promotes cell proliferation in hormone-dependent breast cancer cells
532 by activating the ESR1 (Thongprakaisang et al. 2013; Mesnage et al. 2017). Similarly, we and
533 other authors have shown that glyphosate and GBHs can modulate the expression of both ER
534 subtypes in the rat uterus (Varayoud et al. 2017) as well as induce ESR1 and ESR2 protein
535 expression in T47D breast cancer cells (Thongprakaisang et al. 2013). In the present study, we
536 observed that postnatal exposure to GBH enhanced ESR1 and ESR2 expression in the rat uterus
537 after E2 treatment. However, we found some differences in the expression pattern of both ERs
538 at mRNA level. The results indicate different sensitivity between *Esr1* and *Esr2* in response to
539 GBH exposure. While downregulation of *Esr1* caused by GBH exposure was similar from that
540 induced by E2 treatment alone, the enhancement of *Esr2* with GBH was lower than E2
541 treatment alone. Similar to our findings, other study showed that the expression of both ERs in
542 human breast cancer cells *in vitro* was differentially modulated after glyphosate treatment
543 (Thongprakaisang et al. 2013). Taken together, the above-mentioned results contribute to a
544 growing body of evidence supporting the potential interaction of GBHs with the estrogen
545 signaling pathways.

546

547 In the last years, one matter of concern has focused on toxicological properties of chemicals
548 used in commercial formulations of GBHs. GBHs are commercialized in the form of mixtures
549 consisting of glyphosate (active principle) and different co-formulants (i.e. solvents, penetrating
550 agents, preservatives, surfactants) whose identity is frequently undisclosed because it is
551 considered to be confidential commercial information. Several studies comparing GBHs, co-
552 formulants and glyphosate effects have revealed that GBHs are more toxic than the active
553 ingredient, suggesting that co-formulants can also be a source of toxicity (Defarge et al. 2016;
554 Mesnage & Antoniou 2018). Considering these points, under the condition of our experiment
555 we cannot ascribe the increased uterine sensitivity to E2 only to glyphosate, to co-formulants or
556 both acting together. Further research is required to fully understand glyphosate and co-
557 formulants contribution to GBHs effects.

558

559 It is well-recognized that estrogens and their receptors have a profound involvement in uterine
560 pathology. Consistent with this concept, increased sensitivity to E2 may predispose to
561 endometrial dysfunction and/or disease development. Some EDCs have been demonstrated to
562 increase E2 responses in hormone-sensitive tissues and promote the development of tissue
563 lesions and carcinogenesis later in life. Perinatal exposure of female rodents to low doses of
564 BPA has been shown to enhance sensitivity of the mammary gland to E2 at puberty (Muñoz-de-
565 Toro et al. 2005; Wadia et al. 2007) and to elicit the development of mammary carcinomas
566 during adulthood in the absence of any additional treatment (Acevedo et al. 2013). Other
567 authors have also reported that developmental exposure to BPA alters uterine and prostate gland
568 response to E2 in aged rats, causing increased incidence of pre-neoplastic lesions in the uterus
569 (Vigizzi et al. 2015, 2016) and intraepithelial neoplasia in the prostate (Ho et al. 2006).
570 Similarly, mice postnatally exposed to low doses of DES with an enhanced uterine response to
571 E2 at puberty (Newbold et al. 2004) developed uterine endometrial adenocarcinoma later in life
572 (Newbold et al. 1990). In the present study, we found that early postnatal exposure to GBH
573 altered uterine response to E2 in prepubertal rats. These results may indicate that postnatal GBH
574 exposure could predispose to endometrial pathologies such as carcinogenesis in aged rats.

575

576 In conclusion, our results showed that postnatal exposure to GBH enhances the sensitivity of the
577 rat uterus to estradiol, and induces histomorphologic and molecular changes associated with
578 uterine hyperplasia. GBH treatment induced morphological, cellular and molecular alterations in
579 the rat uterus that were evidenced by the development of uterine hyperplasia, increase in E2-
580 induced cell proliferation, induction of both ESR1 and ESR2, and deregulation of Wnt/ β -catenin
581 signaling pathway. Currently, further experiments are underway to investigate whether
582 subsequent estrogen exposure associated with puberty could induce benign and/or malignant
583 tumors of female reproductive tissues in postnatally GBH-exposed rats.

584

585 **Declaration of interest**

586 The authors declare that there is no conflict of interest that could be perceived as prejudicing the
587 impartiality of the research reported.

588

589 **Funding**

590 This work was supported by grants from Consejo Nacional de Investigaciones Científicas y
591 Técnicas (PIP 2015 N°11220150100338CO), and Agencia Nacional de Promoción Científica y
592 Tecnológica (ANPCyT, PICT 2014 N°1522), Argentina. These funding sources were not
593 involved in the study design, sample collection, analysis or interpretation of the data, the writing
594 of the report, or the decision to submit the article for publication. MGS is a fellow of the
595 CONICET. MMM, EHL and JV are Career Investigators of the CONICET.

596

597 **Acknowledgments**

598 We sincerely thank Juan Grant and Juan C. Villarreal (UNL) for their technical assistance and
599 animal care. We are most grateful to MD. Adriana Albertengo for technical assistance in
600 histological analysis.

601

602 **References**

- 603 Acevedo N, Davis B, Schaeberle CM, Sonnenschein C & Soto AM 2013 Perinatally
604 administered bisphenol A as a potential mammary gland carcinogen in rats. *Environ Health*
605 *Perspect* **121** 1040-1046.
- 606 Arregui MC, Lenardon A, Sanchez D, Maitre MI, Scotta R & Enrique S 2004 Monitoring
607 glyphosate residues in transgenic glyphosate-resistant soybean. *Pest Manag Sci* **60** 163-166.
- 608 Bai SH & Ogbourne SM 2016 Glyphosate: environmental contamination, toxicity and
609 potential risks to human health via food contamination. *Environ Sci Pollut Res Int* **23** 18988-
610 19001.
- 611 Benbrook CM 2016 Trends in glyphosate herbicide use in the United States and globally.
612 *Environ Sci Eur* **28** 016-0070.
- 613 Bonansea RI, Filippi I, Wunderlin DA, Marino DJG & Ame MV 2017 The fate of
614 glyphosate and AMPA in a freshwater endorheic basin: an ecotoxicological risk assessment.
615 *Toxics* **6** 1-13.
- 616 Bottner M & Wuttke W 2005 Chronic treatment with low doses of estradiol affects
617 pituitary and thyroid function in young and middle-aged ovariectomized rats. *Biogerontology* **6**
618 261-269.
- 619 Brembeck FH, Rosario M & Birchmeier W 2006 Balancing cell adhesion and Wnt
620 signaling, the key role of beta-catenin. *Curr Opin Genet Dev* **16** 51-59.
- 621 Bulun SE, Cheng YH, Pavone ME, Xue Q, Attar E, Trukhacheva E, Tokunaga H,
622 Utsunomiya H, Yin P, Luo X, et al. 2010 Estrogen receptor-beta, estrogen receptor-alpha, and
623 progesterone resistance in endometriosis. *Semin Reprod Med* **28** 36-43.
- 624 Bulun SE, Lin Z, Imir G, Amin S, Demura M, Yilmaz B, Martin R, Utsunomiya H, Thung
625 S, Gurates B, et al. 2005 Regulation of aromatase expression in estrogen-responsive breast and
626 uterine disease: from bench to treatment. *Pharmacol Rev* **57** 359-383.
- 627 Cassault-Meyer E, Gress S, Seralini GE & Galeraud-Denis I 2014 An acute exposure to
628 glyphosate-based herbicide alters aromatase levels in testis and sperm nuclear quality. *Environ*
629 *Toxicol Pharmacol* **38** 131-140.

- 630 Cikos S, Bukovská A & Koppel J 2007 Relative quantification of mRNA: comparison of
631 methods currently used for real-time PCR data analysis. *BMC Mol Biol* **8** 113.
- 632 Cohn BA, La Merrill M, Krigbaum NY, Yeh G, Park JS, Zimmermann L & Cirillo PM
633 2015 DDT exposure in utero and breast cancer. *J Clin Endocrinol Metab* **100** 2865-2872.
- 634 Chuffa LG, Lupi-Junior LA, Costa AB, Amorim JP & Seiva FR 2017 The role of sex
635 hormones and steroid receptors on female reproductive cancers. *Steroids* **118** 93-108.
- 636 Defarge N, Takacs E, Lozano VL, Mesnage R, Spiroux de Vendomois J, Seralini GE &
637 Szekacs A 2016 Co-formulants in glyphosate-based herbicides disrupt aromatase activity in
638 human cells below toxic levels. *Int J Environ Res Public Health* **13** 1-17.
- 639 Dixon D, Alison R, Bach U, Colman K, Foley GL, Harleman JH, Haworth R, Herbert R,
640 Heuser A, Long G, et al. 2014 Nonproliferative and proliferative lesions of the rat and mouse
641 female reproductive system. *J Toxicol Pathol* **27** 1-107.
- 642 Durando M, Kass L, Piva J, Sonnenschein C, Soto AM, Luque EH & Muñoz-de-Toro M
643 2007 Prenatal bisphenol A exposure induces preneoplastic lesions in the mammary gland in
644 Wistar rats. *Environ Health Perspect* **115** 80-86.
- 645 EFSA 2015 Conclusion on the peer review of the pesticide risk assessment of the active
646 substance glyphosate. *EFSA Journal* **13** 4302.
- 647 Florescu MM, Pirici D, Simionescu CE, Stepan AE, Margaritescu C, Tudorache S &
648 Ciurea RN 2016 E-cadherin and B-catenin immunoexpression in endometrioid endometrial
649 carcinoma. *Rom J Morphol Embryol* **57** 1235-1240.
- 650 Gaetje R, Holtrich U, Karn T, Cikrit E, Engels K, Rody A & Kaufmann M 2007
651 Characterization of WNT7A expression in human endometrium and endometriotic lesions.
652 *Fertil Steril* **88** 1534-1540
- 653 Gasnier C, Dumont C, Benachour N, Clair E, Chagnon MC & Seralini GE 2009
654 Glyphosate-based herbicides are toxic and endocrine disruptors in human cell lines. *Toxicology*
655 **262** 184-191.
- 656 Guarino M, Rubino B & Ballabio G 2007 The role of epithelial-mesenchymal transition in
657 cancer pathology. *Pathology* **39** 305-318.

- 658 Guerrero Schimpf M, Milesi MM, Ingaramo PI, Luque EH & Varayoud J 2017 Neonatal
659 exposure to a glyphosate based herbicide alters the development of the rat uterus. *Toxicology*
660 **376** 2-14.
- 661 Gundersen HJ, Bendtsen TF, Korbo L, Marcussen N, Moller A, Nielsen K, Nyengaard JR,
662 Pakkenberg B, Sorensen FB, Vesterby A, et al. 1988 Some new, simple and efficient
663 stereological methods and their use in pathological research and diagnosis. *Apmis* **96** 379-394.
- 664 Gunin AG, Emelianov VU, Mironkin IU, Morozov MP & Tolmachev AS 2004 Lithium
665 treatment enhances estradiol-induced proliferation and hyperplasia formation in the uterus of
666 mice. *Eur J Obstet Gynecol Reprod Biol* **114** 83-91.
- 667 Hapangama DK, Kamal AM & Bulmer JN 2015 Estrogen receptor beta: the guardian of the
668 endometrium. *Hum Reprod Update* **21** 174-193.
- 669 Hendry WJ 3rd, Zheng X, Leavitt WW, Branham WS & Sheehan DM 1997 Endometrial
670 hyperplasia and apoptosis following neonatal diethylstilbestrol exposure and subsequent
671 estrogen stimulation in both host and transplanted hamster uteri. *Cancer Res* **57** 1903-1908.
- 672 Ho SM, Tang WY, Belmonte de Frausto J & Prins GS 2006 Developmental exposure to
673 estradiol and bisphenol A increases susceptibility to prostate carcinogenesis and epigenetically
674 regulates phosphodiesterase type 4 variant 4. *Cancer Res* **66** 5624-5632.
- 675 IARC working group on the evaluation of carcinogenic risks to humans 2015 Some
676 organophosphate insecticides and herbicides. *IARC Monographs* **112** 321-412.
- 677 Ingaramo PI, Varayoud J, Milesi MM, Guerrero Schimpf M, Alarcon R, Muñoz-de-Toro M
678 & Luque EH 2017 Neonatal exposure to a glyphosate-based herbicide alters uterine
679 decidualization in rats. *Reprod Toxicol* **73** 87-95.
- 680 Ingaramo PI, Varayoud J, Milesi MM, Guerrero Schimpf M, Muñoz-de-Toro M & Luque
681 EH 2016 Effects of neonatal exposure to a glyphosate-based herbicide on female rat
682 reproduction. *Reproduction* **152** 403-415.
- 683 Kass L, Altamirano GA, Bosquiaz VL, Luque EH & Muñoz-de-Toro M 2012 Perinatal
684 exposure to xenoestrogens impairs mammary gland differentiation and modifies milk
685 composition in Wistar rats. *Reprod Toxicol* **33** 390-400.

686 Leavitt WW, Evans RW & Hendry WJ 3rd 1981 Etiology of DES-induced uterine tumors
687 in the Syrian hamster. *Adv Exp Med Biol* **138** 63-86.

688 Lerro CC, Koutros S, Andreotti G, Friesen MC, Alavanja MC, Blair A, Hoppin JA, Sandler
689 DP, Lubin JH, Ma X, et al. 2015 Organophosphate insecticide use and cancer incidence among
690 spouses of pesticide applicators in the Agricultural Health Study. *Occup Environ Med* **72** 736-
691 744.

692 Liu Y, Meng F, Xu Y, Yang S, Xiao M, Chen X & Lou G 2013 Overexpression of Wnt7a
693 is associated with tumor progression and unfavorable prognosis in endometrial cancer. *Int J*
694 *Gynecol Cancer* **23** 304-311.

695 Ma R & Sassoon DA 2006 PCBs exert an estrogenic effect through repression of the
696 Wnt7a signaling pathway in the female reproductive tract. *Environ Health Perspect* **114** 898-
697 904.

698 Mesnage R & Antoniou MN 2018 Ignoring adjuvant toxicity falsifies the safety profile of
699 commercial pesticides. *Front Public Health* **5** 1-8.

700 Mesnage R, Phedonos A, Biserni M, Arno M, Balu S, Corton JC, Ugarte R & Antoniou
701 MN 2017 Evaluation of estrogen receptor alpha activation by glyphosate-based herbicide
702 constituents. *Food Chem Toxicol* **108** 30-42.

703 Milesi MM, Alarcon R, Ramos JG, Muñoz-de-Toro M, Luque EH & Varayoud J 2015
704 Neonatal exposure to low doses of endosulfan induces implantation failure and disrupts uterine
705 functional differentiation at the pre-implantation period in rats. *Mol Cell Endocrinol* **401** 248-
706 259.

707 Milesi MM, Varayoud J, Bosquiaz VL, Muñoz-de-Toro M & Luque EH 2012 Neonatal
708 exposure to low doses of endosulfan disrupts the expression of proteins regulating uterine
709 development and differentiation. *Reprod Toxicol* **33** 85-93.

710 Monje L, Varayoud J, Muñoz-de-Toro M, Luque EH & Ramos JG 2009 Neonatal exposure
711 to bisphenol A alters estrogen-dependent mechanisms governing sexual behavior in the adult
712 female rat. *Reprod Toxicol* **28** 435-442.

- 713 Muñoz-de-Toro M, Markey CM, Wadia PR, Luque EH, Rubin BS, Sonnenschein C & Soto
714 AM 2005 Perinatal exposure to bisphenol-A alters peripubertal mammary gland development in
715 mice. *Endocrinology* **146** 4138-4147.
- 716 Muñoz-de-Toro MM, Maffini MV, Kass L & Luque EH 1998 Proliferative activity and
717 steroid hormone receptor status in male breast carcinoma. *J Steroid Biochem Mol Biol* **67** 333-
718 339.
- 719 Mutter GL, Zaino RJ, Baak JP, Bentley RC & Robboy SJ 2007 Benign endometrial
720 hyperplasia sequence and endometrial intraepithelial neoplasia. *Int J Gynecol Pathol* **26** 103-
721 114.
- 722 Nakajima T, Tanimoto Y, Tanaka M, Chambon P, Watanabe H, Iguchi T & Sato T 2015
723 Neonatal estrogen receptor beta is important in the permanent inhibition of epithelial cell
724 proliferation in the mouse uterus. *Endocrinology* **156** 3317-3328.
- 725 Newbold RR, Bullock BC & McLachlan JA 1990 Uterine adenocarcinoma in mice
726 following developmental treatment with estrogens: a model for hormonal carcinogenesis.
727 *Cancer Res* **50** 7677-7681.
- 728 Newbold RR, Jefferson WN & Padilla-Banks E 2007 Long-term adverse effects of
729 neonatal exposure to bisphenol A on the murine female reproductive tract. *Reprod Toxicol* **24**
730 253-258.
- 731 Newbold RR, Jefferson WN, Padilla-Banks E & Haseman J 2004 Developmental exposure
732 to diethylstilbestrol (DES) alters uterine response to estrogens in prepubescent mice: low versus
733 high dose effects. *Reprod Toxicol* **18** 399-406.
- 734 Newbold RR, Jefferson WN, Padilla-Burgos E & Bullock BC 1997 Uterine carcinoma in
735 mice treated neonatally with tamoxifen. *Carcinogenesis* **18** 2293-2298.
- 736 Niemann L, Sieke C, Pfeil R & Solecki R 2015 A critical review of glyphosate findings in
737 human urine samples and comparison with the exposure of operators and consumers. *Journal*
738 *für Verbraucherschutz und Lebensmittelsicherheit* **10** 3-12.

- 739 Perego MC, Schutz LF, Caloni F, Cortinovis C, Albonico M & Spicer LJ 2017 Evidence
740 for direct effects of glyphosate on ovarian function: glyphosate influences steroidogenesis and
741 proliferation of bovine granulosa but not theca cells in vitro. *J Appl Toxicol* **37** 692-698.
- 742 Peruzzo PJ, Porta AA & Ronco AE 2008 Levels of glyphosate in surface waters, sediments
743 and soils associated with direct sowing soybean cultivation in north pampasic region of
744 Argentina. *Environ Pollut* **156** 61-66.
- 745 Portier CJ, Armstrong BK, Baguley BC, Baur X, Belyaev I, Belle R, Belpoggi F, Biggeri
746 A, Bosland MC, Bruzzi P, et al. 2016 Differences in the carcinogenic evaluation of glyphosate
747 between the International Agency for Research on Cancer (IARC) and the European Food
748 Safety Authority (EFSA). *J Epidemiol Community Health* **70** 741-745.
- 749 Preisler-Mashek MT, Solodin N, Stark BL, Tyrivier MK & Alarid ET 2002 Ligand-specific
750 regulation of proteasome-mediated proteolysis of estrogen receptor-alpha. *Am J Physiol*
751 *Endocrinol Metab* **282** 891-898.
- 752 Primost JE, Marino DJG, Aparicio VC, Costa JL & Carriquiriborde P 2017 Glyphosate and
753 AMPA, "pseudo-persistent" pollutants under real-world agricultural management practices in
754 the Mesopotamic Pampas agroecosystem, Argentina. *Environ Pollut* **229** 771-779.
- 755 Ramos JG, Varayoud J, Bosquiaz VL, Luque EH & Muñoz-de-Toro M 2002 Cellular
756 turnover in the rat uterine cervix and its relationship to estrogen and progesterone receptor
757 dynamics. *Biol Reprod* **67** 735-742.
- 758 Ramos JG, Varayoud J, Monje L, Moreno-Piovan G, Muñoz-de-Toro M & Luque EH
759 2007 Diethylstilbestrol alters the population dynamic of neural precursor cells in the neonatal
760 male rat dentate gyrus. *Brain Res Bull* **71** 619-627.
- 761 Ramos JG, Varayoud J, Sonnenschein C, Soto AM, Muñoz-De-Toro M & Luque EH 2001
762 Prenatal exposure to low doses of bisphenol A alters the periductal stroma and glandular cell
763 function in the rat ventral prostate. *Biol Reprod* **65** 1271-1277.
- 764 Richard S, Moslemi S, Sipahutar H, Benachour N & Seralini GE 2005 Differential effects
765 of glyphosate and roundup on human placental cells and aromatase. *Environ Health Perspect*
766 **113** 716-720.

- 767 Rivera OE, Varayoud J, Rodriguez HA, Muñoz-de-Toro M & Luque EH 2011 Neonatal
768 exposure to bisphenol A or diethylstilbestrol alters the ovarian follicular dynamics in the lamb.
769 *Reprod Toxicol* **32** 304-312.
- 770 Rivera OE, Varayoud J, Rodriguez HA, Santamaria CG, Bosquiazazo VL, Osti M,
771 Belmonte NM, Muñoz-de-Toro M & Luque EH 2015 Neonatal exposure to xenoestrogens
772 impairs the ovarian response to gonadotropin treatment in lambs. *Reproduction* **149** 645-655.
- 773 Rodrigues NR & de Souza APF 2017 Occurrence of glyphosate and AMPA residues in
774 soy-based infant formula sold in Brazil. *Food Addit Contam Part A Chem Anal Control Expo*
775 *Risk Assess* **20** 1-8.
- 776 Saegusa M, Hashimura M, Yoshida T & Okayasu I 2001 Beta-catenin mutations and
777 aberrant nuclear expression during endometrial tumorigenesis. *Br J Cancer* **84** 209-217.
- 778 Sanderson PA, Critchley HO, Williams AR, Arends MJ & Saunders PT 2017 New
779 concepts for an old problem: the diagnosis of endometrial hyperplasia. *Hum Reprod Update* **23**
780 232-254.
- 781 Smith MS, Freeman ME & Neill JD 1975 The control of progesterone secretion during the
782 estrous cycle and early pseudopregnancy in the rat: prolactin, gonadotropin and steroid levels
783 associated with rescue of the corpus luteum of pseudopregnancy. *Endocrinology* **96** 219-226.
- 784 Solomon KR 2016 Glyphosate in the general population and in applicators: a critical
785 review of studies on exposures. *Crit Rev Toxicol* **46** 21-27.
- 786 Tarazona JV, Court-Marques D, Tiramani M, Reich H, Pfeil R, Istace F & Crivellente F
787 2017 Glyphosate toxicity and carcinogenicity: a review of the scientific basis of the European
788 Union assessment and its differences with IARC. *Arch Toxicol* **91** 2723-2743.
- 789 Test Biotech 2013 High levels of residues from spraying with glyphosate found in
790 soybeans in Argentina. <http://www.testbiotech.org/en/node/926> Accessed 28 June 2018.
- 791 Thongprakaisang S, Thiantanawat A, Rangkadilok N, Suriyo T & Satayavivad J 2013
792 Glyphosate induces human breast cancer cells growth via estrogen receptors. *Food Chem*
793 *Toxicol* **59** 129-136.

- 794 Tica AA, Tica OS, Georgescu CV, Pirici D, Bogdan M, Ciurea T, Mogoanta SS,
795 Georgescu CC, Comanescu AC, Balseanu TA, et al. 2016 GPER and ER alpha expression in
796 abnormal endometrial proliferations. *Rom J Morphol Embryol* **57** 413-418.
- 797 US EPA 2017 Glyphosate. Dietary Exposure Analysis in Support of Registration Review.
798 <https://www.regulations.gov/document?D=EPA-HQ-OPP-2009-0361-0071>, accessed 28 June
799 2018.
- 800 US EPA 2015 EDSP: weight of evidence analysis of potential interaction with the estrogen,
801 androgen or thyroid pathways. Chemical: glyphosate. Office of Pesticide Programs US EPA.
802 [https://www.epa.gov/endocrine-disruption/endocrine-disruptor-screening-program-tier-1-](https://www.epa.gov/endocrine-disruption/endocrine-disruptor-screening-program-tier-1-screening-determinations-and)
803 [screening-determinations-and](https://www.epa.gov/endocrine-disruption/endocrine-disruptor-screening-program-tier-1-screening-determinations-and), accessed January 26, 2018.
- 804 Van Bruggen AHC, He MM, Shin K, Mai V, Jeong KC, Finckh MR & Morris JG, Jr. 2018
805 Environmental and health effects of the herbicide glyphosate. *Sci Total Environ* **617** 255-268.
- 806 Van der Horst PH, Wang Y, Van der Zee M, Burger CW & Blok LJ 2012 Interaction
807 between sex hormones and WNT/B-catenin signal transduction in endometrial physiology and
808 disease. *Mol Cell Endocrinol* **358** 176-184.
- 809 Vandenberg LN, Blumberg B, Antoniou MN, Benbrook CM, Carroll L, Colborn T, Everett
810 LG, Hansen M, Landrigan PJ, Lanphear BP, et al. 2017 Is it time to reassess current safety
811 standards for glyphosate-based herbicides? *J Epidemiol Community Health* **71** 613-618.
- 812 Varayoud J, Durando M, Ramos JG, Milesi MM, Ingaramo PI, Muñoz-de-Toro M &
813 Luque EH 2017 Effects of a glyphosate-based herbicide on the uterus of adult ovariectomized
814 rats. *Environ Toxicol* **32** 1191-1201.
- 815 Varayoud J, Ramos JG, Bosquiazzo VL, Muñoz-de-Toro M & Luque EH 2008
816 Developmental exposure to bisphenol A impairs the uterine response to ovarian steroids in the
817 adult. *Endocrinology* **149** 5848-5860.
- 818 Varayoud J, Ramos JG, Muñoz-de-Toro M & Luque EH 2014 Long-lasting effects of
819 neonatal bisphenol A exposure on the implantation process. *Vitam Horm* **94** 253-275.

- 820 Viguzzi L, Bosquiazzo VL, Kass L, Ramos JG, Muñoz-de-Toro M & Luque EH 2015
821 Developmental exposure to bisphenol A alters the differentiation and functional response of the
822 adult rat uterus to estrogen treatment. *Reprod Toxicol* **52** 83-92.
- 823 Viguzzi L, Ramos JG, Kass L, Tschopp MV, Muñoz-de-Toro M, Luque EH & Bosquiazzo
824 VL 2016 A deregulated expression of estrogen-target genes is associated with an altered
825 response to estradiol in aged rats perinatally exposed to bisphenol A. *Mol Cell Endocrinol* **426**
826 33-42.
- 827 Wada-Hiraike O, Hiraike H, Okinaga H, Imamov O, Barros RP, Morani A, Omoto Y,
828 Warner M & Gustafsson JA 2006 Role of estrogen receptor beta in uterine stroma and
829 epithelium: Insights from estrogen receptor beta-/- mice. *Proc Natl Acad Sci U S A* **103** 18350-
830 18355.
- 831 Wadia PR, Vandenberg LN, Schaeberle CM, Rubin BS, Sonnenschein C & Soto AM 2007
832 Perinatal bisphenol A exposure increases estrogen sensitivity of the mammary gland in diverse
833 mouse strains. *Environ Health Perspect* **115** 592-598.
- 834 Wang Y, Van der Zee M, Fodde R & Blok LJ 2010 Wnt/B-catenin and sex hormone
835 signaling in endometrial homeostasis and cancer. *Oncotarget* **1** 674-684.
- 836 Weibel ER 1969 Stereological principles for morphometry in electron microscopic
837 cytology. *Int Rev Cytol* **26** 235-302.
- 838 Winuthayanon W, Hewitt SC & Korach KS 2014 Uterine epithelial cell estrogen receptor
839 alpha-dependent and -independent genomic profiles that underlie estrogen responses in mice.
840 *Biol Reprod* **91** 1-10.
- 841 Winuthayanon W, Hewitt SC, Orvis GD, Behringer RR & Korach KS 2010 Uterine
842 epithelial estrogen receptor alpha is dispensable for proliferation but essential for complete
843 biological and biochemical responses. *Proc Natl Acad Sci USA* **107** 19272-19277.
- 844 Yamashita S 2006 Expression of estrogen-regulated genes during development in the
845 mouse uterus exposed to diethylstilbestrol neonatally. *Curr Pharm Des* **12** 1505-1520.

846 Yoshizawa K, Brix AE, Sells DM, Jokinen MP, Wyde M, Orzech DP, Kissling GE, Walker
847 NJ & Nyska A 2009 Reproductive lesions in female Harlan Sprague-Dawley rats following
848 two-year oral treatment with dioxin and dioxin-like compounds. *Toxicol Pathol* **37** 921-937.

849 Zoller O, Rhyn P, Rupp H, Zarn JA & Geiser C 2017 Glyphosate residues in Swiss market
850 foods: monitoring and risk evaluation. *Food Addit Contam Part B Surveill* **28** doi
851 10.1080/19393210.2017.1419509.

852

1 **Figure legends**

2 **Fig. 1.** Schematic representation of the experimental protocol used to investigate the effects of
3 postnatal exposure to a glyphosate-based herbicide (GBH) on uterine sensitivity to 17 β -estradiol
4 (E2). C, control; OVX, ovariectomy; sc, subcutaneous; V, vehicle (corn oil).

5 **Fig. 2.** Uterine histology in OVX prepubertal control rats following vehicle treatment
6 (C+OVX+V) or E2 stimulation (C+OVX+E2). Representative images of longitudinal sections
7 of uterine horns of C+OVX+V and C+OVX+E2 animals stained with H&E at low (A and B)
8 and high (C and D) magnification. LE, luminal epithelium; GE, glandular epithelium; SS,
9 subepithelial stroma; MM, myometrium. Scale bar: (A and B) 100 μ m; (C and D) 50 μ m.

10 **Fig. 3.** Uterine responsiveness to E2 stimulation in OVX prepubertal rats: expression of the
11 estrogen receptors alpha (ESR1) and beta (ESR2), and cell proliferation. (A and B) Relative
12 mRNA levels of *Esr1* and *Esr2* in vehicle- (C+OVX+V) and E2-treated (C+OVX+E2) animals
13 were measured by qRT-PCR. Fold expression from C+OVX+V values was calculated by the
14 relative standard curve method. C+OVX+V values were assigned to a reference level of 1 and
15 values are given as mean $\pm$ SEM of two independent determinations. Significant effects are
16 depicted with asterisks (* $p < 0.05$). (C) Quantification of ESR1 and ESR2 protein
17 immunostaining is expressed as the integrated optical density (IOD). Cell proliferation is shown
18 as a percentage of Ki67-positive cells. Significant effects are depicted with asterisks (* $p < 0.05$;
19 ** $p < 0.01$). (D-I) Representative images of protein expression of the ESR1 (D and E) and ESR2
20 (F and G), and cellular proliferative responses by Ki67 immunodetection (H and I) in
21 C+OVX+V and C+OVX+E2 groups. LE, luminal epithelium; GE, glandular epithelium; SS,
22 subepithelial stroma. Scale bar: 50 μ m.

23 **Fig. 4.** Uterine histological lesions in postnatally GBH-exposed animals after E2 stimulation at
24 low and high magnification. (A and D) Normal uterine histology in OVX control rats after E2
25 treatment (C+OVX+E2). (B, C, E and F) Representative H&E stained images of postnatally
26 GBH-exposed rats after E2 stimulation (GBH+OVX+E2) demonstrating histological findings.
27 Abnormal endometrial glands (arrows) and areas of endometrium with stratification of the
28 luminal epithelium (asterisk), hypochromic nuclei (asterisk) and mitotic figures (arrowheads)

are indicated. Uterine areas in the insets are magnified in the lower panel. LE, luminal epithelium; GE, glandular epithelium; SS, subepithelial stroma; MM, myometrium. Scale bar: (A, B and C) 50 μ m; (D, E and F) 10 μ m.

Fig. 5. Effects of GBH postnatal exposure on E2-induced cell proliferation. (A) Ki67 protein expression was quantified in luminal and glandular epithelial uterine cells. Values in bar graphs are the mean $\pm$ SEM. The asterisk indicates statistical significance compared with the control (* $p < 0.05$). (B) Representative images of immunohistochemical detection of Ki67 on uterine sections. The asterisk indicates the increase of Ki67 expression. LE, luminal epithelium; GE, glandular epithelium; SS, subepithelial stroma. Scale bar: 50 μ m.

Fig. 6. Effects of GBH postnatal exposure on uterine expression of the estrogen receptors alpha (ESR1) and beta (ESR2). (A and B) Relative mRNA levels of *Esr1* (A) and *Esr2* (B) in control and postnatally GBH-exposed animals were measured by qRT-PCR. Fold expression from C+OVX+E2 values was calculated by the relative standard curve method. C+OVX+E2 values were assigned to a reference level of 1 and values are given as mean $\pm$ SEM of two independent determinations. Significant effects are depicted with asterisks (* $p < 0.05$). (C) Quantification of ESR1 and ESR2 protein immunostaining in uterine regions is expressed as the integrated optical density (IOD). Values in bar graphs are the mean $\pm$ SEM. Asterisks indicate statistical significance compared with the control (* $p < 0.05$). (D) Representative images of immunohistochemical staining for ESR1 and ESR2 on uterine sections. Significant effects are depicted with asterisks. LE, luminal epithelium; GE, glandular epithelium; SS, subepithelial stroma. Scale bar: 50 μ m.

Fig. 7. Effects of GBH postnatal exposure on the expression of E2-modulated targets that have been implicated in altered uterine E2 responsiveness and neoplasia. (A and B) Relative mRNA levels of *Wnt7a* (A) and *Ctnnb1* (B) in control and postnatally GBH-exposed animals were measured by qRT-PCR. Fold expression from C+OVX+E2 values was calculated by the relative standard curve method. C+OVX+E2 values were assigned to a reference level of 1 and values are given as mean $\pm$ SEM of two independent determinations. Significant effects are depicted with asterisks (* $p < 0.05$). (C) Quantification of WNT7A and β -catenin protein immunostaining

57 in uterine regions is expressed as the integrated optical density (IOD). Values in bar graphs are
58 the mean $\pm$ SEM. Significant effects are depicted with asterisks (* $p < 0.05$). (D) Representative
59 images of immunohistochemical staining for WNT7A and β -catenin on uterine sections.
60 Significant effects are depicted with asterisks. LE, luminal epithelium; GE, glandular
61 epithelium; SS, subepithelial stroma. Scale bar: 50 μ m.
62

Table 1

Primers used for qRT-PCR

Gene name	Primer Sense (5' → 3')	Primer Antisense (5' → 3')	Product size (bp)
<i>Esr1</i>	ACTACCTGGAGAACGAGCCC	CCTTGGCAGACTCCATGATC	153
<i>Esr2</i>	TTCTGGGCACCTGTCTCCTT	TAACAGGGCTGGCACAACCTG	166
<i>Ctnnb1</i>	GAGCACATCAGGACACCCAGC	GAGGATGTGGAGAGCCCCAGT	116
<i>Wnt7a</i>	CTTACACAATAACGAGGCAGGC	TCTCGGAATTGTGGCAGTGT	126
<i>Rpl19</i>	AGCCTGTGACTGTCCATTCC	TGGCAGTACCCTTCCTCTTC	99

Esr1, estrogen receptor alpha; *Esr2*, estrogen receptor beta; *Ctnnb1*, beta-catenin; *Rpl19*, ribosomal protein L19; *Wnt7a*, wingless-type MMTV integration site family, member 7A.

Table 2

Antibodies used for immunohistochemistry

Antibodies	Dilution	Supplier
<i>Primary</i>		
Anti-Ki67 (clone MIB-5)	1/50	Dako Corp. (Carpinteria, CA, USA)
Anti-ESR1 (clone 6F-11)	1/200	Novocastra (Newcastle upon Tyne, UK)
Anti-ESR2 (clone EMR02)	1/50	Novocastra (Newcastle upon Tyne, UK)
Anti-WNT7A	1/800	Generated and validated in our laboratory (Vigazzi et al. 2016)
Anti- β -catenin (sc-7963)	1/800	Santa Cruz Biotechnology Inc. (Santa Cruz, CA, USA)
<i>Secondary</i>		
Anti-mouse (B8774)	1/100	Sigma-Aldrich (St. Louis, MO, USA)
Anti-rabbit (B8895)	1/200	Sigma-Aldrich (St. Louis, MO, USA)
ESR1, estrogen receptor alpha; ESR2, estrogen receptor beta; WNT7A, wingless-type MMTV integration site family, member 7A.		

Table 3Serum levels of 17 β -estradiol (E2)

Treatment group	E2 levels (pg/ml)
C+OVX+V	ND
C+OVX+E2	33.1 $\pm$ 3.3
GBH+OVX+E2	31.6 $\pm$ 2.0

Estradiol levels were measured via RIA. Values are mean $\pm$ SEM (n = 5-8 rats/group).

C, control; E2, 17 β -estradiol (1 mg/ml); GBH, glyphosate-based herbicide (2 mg of glyphosate/kg bw/day); ND, not detected; OVX, ovariectomized; V, vehicle.

Table 4

Morphometric analysis of uterine sections

Parameters	C+OVX+E2	GBH+OVX+E2
Luminal epithelial height (μm)	32.04 ± 4.03	68.10 ± 13.76 *
Glandular epithelial height (μm)	15.30 ± 1.34	15.42 ± 1.16
Endometrial glands (Vv × 100)	4.63 ± 0.21	5.26 ± 0.40
Stromal nuclei density (%)	19.99 ± 0.97	26.04 ± 1.67 **
Stromal thickness (μm)	571.25 ± 39.44	598.12 ± 42.88
Circular myometrium thickness (μm)	159.42 ± 13.85	198.78 ± 26.48
Longitudinal myometrium thickness (μm)	139.89 ± 12.63	173.33 ± 16.69

Values are means ± SEM (n = 7-8 rats/group).

C, control; E2, 17β-estradiol (1 mg/ml); GBH, glyphosate-based herbicide (2 mg of glyphosate/kg b.w/day); OVX, ovariectomized; Vv, volume fraction. * p<0.05, **p<0.01 compared with the control group.













